

[Assignment-2]

Name: Imtiaz Hossain

ID : 23101137

Course

code : CSE 370

Faculty : RBR

Initial

Section : 06

[Answer to the question no 1]

a) Insert < 'product A', 4, 'Bellairne', 2 > into PROJECT
constraints violated:

1) Referential Integrity constraint:

The attribute Dnum in PROJECT references Dnumber in DEPARTMENT. Since Dnum = 2 does not exist in DEPARTMENT table (where Dnumber values are 1, 4, and 5), this insertion violates referential integrity.

ways to Enforce constraints:

1) Declare Dnum in the PROJECT table as a foreign key referencing Dnumber in the DEPARTMENT table. Then the DBMS will disallow this operation.

b) Insert <'453453453', 'John', 'M', '1990-12-12', 'Spouse'> into DEPENDENT.

No constraints violated. → The SSN value exists in the Employee table, unique in the DEPENDENT table, so no key constraint.

c) Insert <'Production', 5, '987987989', '2007-10-01'> into DEPARTMENT.

Constraints violated:

1) key constraint: Dnumber = 5 already exists in the DEPARTMENT tuple.

~~ways to enforce constraint:~~

2) Referential Integrity constraint: mgr-SSN = '987987989'

No tuple exists in the EMPLOYEE relation with SSN = '987987989'

ways to enforce constraint:

1) ~~change~~ Declare Dnumber in the DEPARTMENT table as a ~~foreign key~~ primary key. Then the DBMS will disallow the operation.

2) Declare mgr-ssn as a foreign key
referencing ssn in the Employee table.
Then the DBMS will disallow the operation.

d. Delete the EMPLOYEE tuple with ssn = '333445555'

Constraints violated:

1) Referential Integrity: ^{the} ssn = '333445555'
is referenced by

DEPARTMENT, WORKS-ON, and DEPENDENT. If we'll
remove this employee from the Employee
table, then the values in the referencing
tables will not be existing in the referenced
table anymore.

ways to enforce constraint:

1) when declaring the columns in the referencing
table as foreign key also specify what
will happen on delete: restrict → deletion

from referenced table not allowed,
cascade $\rightarrow$ the referencing rows will also
be deleted or set null $\rightarrow$ the value in the
referencing column will be set to null.

e) modify the mgr-ssn and mgr-start date
of the DEPARTMENT tuple with Dnumber = 5
to '123456789' and '2007-10-01', respectively.

No constraints violated.

$\rightarrow$ mgr-ssn is a valid ssn in the
Employee table. The primary key is not
modified, so no key constraint is violated.

The mgr-start-date update is within the
domain of date values.